- Which social media platform (Twitter, Instagram, or Facebook) is the most effective method of advertising in each country? (In this case, consider the total number of lead conversions as a measure of effectiveness).

```
WITH channel_conversions AS (
  SELECT
    m."Country",
    'Twitter' AS channel,
    SUM(a.twitter_ad) AS total_conversions
  FROM marketing_data m
  JOIN ad_data a
    ON m."ID" = a.id
  GROUP BY m."Country"

  UNION ALL

  SELECT
    m."Country",
    'Instagram' AS channel,
    SUM(a.instagram_ad) AS total_conversions
  FROM marketing_data m
  JOIN ad_data a
    ON m."ID" = a.id
  GROUP BY m."Country"

  UNION ALL

  SELECT
    m."Country",
    'Facebook' AS channel,
    SUM(a.facebook_ad) AS total_conversions
  FROM marketing_data m
  JOIN ad_data a
    ON m."ID" = a.id
  GROUP BY m."Country"

  UNION ALL

  SELECT
    m."Country",
```

```
        'Bulkmail' AS channel,
        SUM(a.bulkmail_ad) AS total_conversions
    FROM marketing_data m
    JOIN ad_data a
        ON m."ID" = a.id
    GROUP BY m."Country"

    UNION ALL

    SELECT
        m."Country",
        'Brochure' AS channel,
        SUM(a.brochure_ad) AS total_conversions
    FROM marketing_data m
    JOIN ad_data a
        ON m."ID" = a.id
    GROUP BY m."Country"
),
ranked AS (
    SELECT
        "Country",
        channel,
        total_conversions,
        ROW_NUMBER() OVER (
            PARTITION BY "Country"
            ORDER BY total_conversions DESC, channel
        ) AS rn
    FROM channel_conversions
)
SELECT
    "Country",
    channel AS most_effective_channel,
    total_conversions
FROM ranked
WHERE rn = 1
ORDER BY "Country";
```

	Country text	most_effective_channel text	total_conversions bigint
1	AUS	Instagram	12
2	CA	Twitter	24
3	GER	Twitter	11
4	IND	Bulkmail	13
5	ME	Bulkmail	1
6	SA	Bulkmail	21
7	SP	Instagram	89
8	US	Bulkmail	8

- Which social media platform is the most effective method of advertising based on marital status? (In this case, consider the total number of lead conversions as a measure of effectiveness).

```
WITH platform_conversions AS (
  SELECT
    m."Marital_Status",
    'Twitter' AS platform,
    SUM(a.twitter_ad) AS total_conversions
  FROM marketing_data m
  JOIN ad_data a
    ON m."ID" = a.id
  GROUP BY m."Marital_Status"

  UNION ALL

  SELECT
    m."Marital_Status",
```

```
    'Instagram' AS platform,
    SUM(a.instagram_ad) AS total_conversions
FROM marketing_data m
JOIN ad_data a
    ON m."ID" = a.id
GROUP BY m."Marital_Status"

UNION ALL

SELECT
    m."Marital_Status",
    'Facebook' AS platform,
    SUM(a.facebook_ad) AS total_conversions
FROM marketing_data m
JOIN ad_data a
    ON m."ID" = a.id
GROUP BY m."Marital_Status"

UNION ALL

SELECT
    m."Marital_Status",
    'Bulkmail' AS platform,
    SUM(a.bulkmail_ad) AS total_conversions
FROM marketing_data m
JOIN ad_data a
```

```
    ON m."ID" = a.id
  GROUP BY m."Marital_Status"

  UNION ALL

  SELECT
    m."Marital_Status",
    'Brochure' AS platform,
    SUM(a.brochure_ad) AS total_conversions
  FROM marketing_data m
  JOIN ad_data a
    ON m."ID" = a.id
  GROUP BY m."Marital_Status"
),
ranked AS (
  SELECT
    "Marital_Status",
    platform,
    total_conversions,
    ROW_NUMBER() OVER (
      PARTITION BY "Marital_Status"
      ORDER BY total_conversions DESC
    ) AS rn
  FROM platform_conversions
)
SELECT
```

```
    "Marital_Status",
    platform AS most_effective_platform,
    total_conversions
FROM ranked
WHERE rn = 1
ORDER BY "Marital_Status";
```

	Marital_Status text	most_effective_platform text	total_conversions bigint
1	Absurd	Instagram	1
2	Alone	Bulkmail	1
3	Divorced	Bulkmail	20
4	Married	Instagram	66
5	Single	Bulkmail	38
6	Together	Instagram	44
7	Widow	Twitter	10
8	YOLO	Brochure	0

Which social media platform(s) seem(s) to be the most effective per country? (In this case, assume that purchases were in some way influenced by lead conversions from a campaign). Hint: You'll want to generate the amount spent per product type per country and include a total of the ads for each of the social media platforms. Then, review the output to determine whether there is anything interesting related to the amount spent per product in each country and which social media platforms were used for advertising.

```
SELECT
  m."Country",

  /* Product spend */
  SUM(m."AmtLiq")        AS liquor_spend,
  SUM(m."AmtVege")       AS vegetable_spend,
  SUM(m."AmtNonVeg")     AS nonveg_spend,
  SUM(m."AmtPes")        AS fish_spend,
  SUM(m."AmtChocolates")  AS chocolate_spend,
  SUM(m."AmtComm")       AS commodity_spend,

  /* Social media lead conversions */
  SUM(a.twitter_ad)      AS twitter_conversions,
  SUM(a.instagram_ad)    AS instagram_conversions,
  SUM(a.facebook_ad)     AS facebook_conversions

FROM marketing_data m
JOIN ad_data a
  ON m."ID" = a.id
GROUP BY m."Country"
ORDER BY m."Country";
```

	Country text	liquor_spend numeric	vegetable_spend numeric	nonveg_spend numeric	fish_spend numeric	chocolate_spend numeric	commodity_spend numeric	twitter_conversions bigint	instagram_conversions bigint	facebook_conversions bigint
1	AUS	42752	3689	22328	5546	4129	7132	6	12	7
2	CA	84066	7681	45925	9980	7607	12144	24	21	18
3	GER	36776	2980	20272	4601	2801	5768	11	8	7
4	IND	36236	3788	23729	4818	3221	6014	10	6	7
5	ME	1729	8	817	226	122	220	0	0	0
6	SA	105918	8937	58398	13670	9019	15129	20	21	20
7	SP	336392	28288	178409	40153	30134	46181	87	89	76
8	US	32214	3034	20185	4411	2863	4839	6	5	7